



INTERNSHIP PROJECT REPORT

Afforestation and Deforestation Effects on European Topsoil Carbon, Exploratory Analysis of the German Agricultural National Soil Inventory (BZE) & AI4SoilHealth: Accelerating Collection and Use of Soil Health Information Using AI Technology to Support the Soil Deal for Europe and the EU Soil Observatory

Johann Heinrich von Thünen Institute

Institute of Climate-Smart Agriculture (Institut für Agrarklimaschutz)

Regionalization and Data Science Group

Bundesallee 68, 38116 Braunschweig, Germany

Internship period: 16.03.2026 until 15.06.2026

Submitted by:

Lokesh Goud Dabbi

Study programme: M.Sc. Organic Agriculture and Food Systems

Matriculation number: 1059456

University of Hohenheim, Stuttgart

Supervisor:

Dr. Florian Schneider

Head of the Regionalization and Data Science Group

Institute of Climate-Smart Agriculture

Bundesallee 68, Braunschweig

Declaration of Authenticity

I hereby declare that this internship report is my own work. It is based on the research I carried out and the experience I gained during my internship at the Institute of Climate-Smart Agriculture of the Johann Heinrich von Thünen Institute in Braunschweig between 16 March 2026 and 15 June 2026. All analyses and figures in this report were produced by me in R during the internship. The datasets used are openly available (German Agricultural Soil Inventory) or were provided for research use (LUCAS soil data), and the land use reconstruction was compiled by me from historical aerial imagery. General scientific background on soil organic carbon reflects established knowledge from my studies and from the literature listed in the references.

Stuttgart, 08-10-2026

Place, Date

D Lokesh Goud

Signature (Lokesh Goud Dabbi)

Acknowledgements

I would like to thank my supervisor Dr. Florian Schneider, head of the Regionalization and Data Science group, for his guidance throughout the internship. He always took the time to discuss my results, pointed me to the right data and questions, and explained why an analysis should be done in a certain way rather than just how to do it. I am also grateful to all colleagues at the Institute of Climate-Smart Agriculture for welcoming me so openly, for answering my many questions about soil data, and for trusting me, in my very first week, to help organise the AI4SoilHealth Consortium Meeting in Braunschweig. Finally, I thank the University of Hohenheim and my study programme, M.Sc. Organic Agriculture and Food Systems, for making this practical placement part of my studies.

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List of Abbreviations

Abbreviation	Meaning
AG	Arbeitsgruppe (working group)
BMLEH	Bundesministerium für Landwirtschaft, Ernährung und Heimat (German Federal Ministry of Agriculture, Food and Regional Identity)
BZE	Bodenzustandserhebung (soil inventory)
BZE-LW	Bodenzustandserhebung Landwirtschaft (German Agricultural Soil Inventory)
EU	European Union
GIS	Geographic information system
LUCAS	Land Use and Coverage Area frame Survey (LUCAS-SOIL: its soil module)
QGIS	Open source geographic information system
JRC	Joint Research Centre of the European Commission
SOC	Soil organic carbon
Mg C/ha	Megagrams of carbon per hectare (unit of carbon stock)
g/kg	Grams of carbon per kilogram of soil (unit of carbon concentration)
MAT	Mean annual temperature
MAP	Mean annual precipitation
n	Number of sites in a group
r	Pearson correlation coefficient

Executive Summary

From 16 March to 15 June 2026 I completed a three month internship at the Thünen Institute of Climate-Smart Agriculture in Braunschweig. The Johann Heinrich von Thünen Institute is the German Federal Research Institute for Rural Areas, Forestry and Fisheries and an independent higher federal authority within the portfolio of the German Federal Ministry of Agriculture, Food and Regional Identity (BMLEH). The internship was part of my M.Sc. studies in Organic Agriculture and Food Systems at the University of Hohenheim. I worked in the Regionalization and Data Science group led by Dr. Florian Schneider, where I contributed to the EU project AI4SoilHealth (Grant Agreement No. 101086179).

My work had three parts. In my first week I helped with the planning, logistics and onsite support of the AI4SoilHealth Consortium Meeting, held from 23 to 25 March 2026 in Braunschweig. I then carried out an exploratory analysis of the German Agricultural Soil Inventory with 3,104 sites and learned how soil carbon depends on land use, peat, clay content, bulk density and region. The main part was a study across several countries on how historic land use change affects topsoil organic carbon along a pronounced climate gradient from Spain and France through Baden-Württemberg to Sweden. For about 1,000 LUCAS-SOIL sites I reconstructed the land use history from digitised, georeferenced aerial photographs from 1950 to the present in QGIS, integrated it with published soil carbon measurements, and analysed the combined dataset in a reproducible R pipeline.

The main result is that soils with forest in their history hold about twice as much carbon as soils that stayed cropland or grassland, and that this land use effect remains visible even after climate, the strongest single control, is taken into account. The internship strengthened my skills in data analysis, GIS and soil science, gave me experience in organising an international project meeting, and confirmed my wish to work where soil science and data analysis meet.

1. Introduction

1.1 Background and Motivation

Soil organic carbon links two topics I care about: soil fertility in farming systems and the role of agriculture in climate change. During my master's programme in Organic Agriculture and Food Systems I learned a lot about soil organic matter, nutrient cycles and sustainable land use, but mostly from lectures and textbooks. I wanted to find out how these ideas hold up when they are tested against real measurements from thousands of sites, and how researchers actually turn raw soil data into reliable conclusions.

The Thünen Institute was an ideal place for this. Its Institute of Climate-Smart Agriculture runs the German Agricultural Soil Inventory, which is the basis for national soil carbon reporting, and its Regionalization and Data Science group works with large soil datasets on a daily basis and contributes to the EU project AI4SoilHealth. This gave me the chance to work on a real research question from start to finish, with access to high quality data and to people who know these datasets in detail.

1.2 Structure of this Report

Chapter 2 introduces the Thünen Institute, the Institute of Climate-Smart Agriculture and the working group I was part of. Chapter 3 gives the scientific background on soil organic carbon and on soil monitoring in Germany and Europe. Chapter 4 lists the objectives I set for the internship and Chapter 5 shows how the work was spread over the three months. Chapter 6 is the core of the report and describes my project work: the exploratory analysis of the German soil inventory, the main study on afforestation and deforestation in Europe with all results, and my contribution to the EU project AI4SoilHealth, including the organisation of its consortium meeting. Chapters 7 to 10 cover the skills I gained, the challenges I faced, my personal reflection and the link to my studies. Chapter 11 concludes the report, followed by the references and a glossary.

Internship Details at a Glance

Host institution	Johann Heinrich von Thünen Institute, Federal Research Institute for Rural Areas, Forestry and Fisheries (within the portfolio of the BMLEH)
Institute	Institute of Climate-Smart Agriculture (Institut für Agrarklimaschutz)
Working group	Regionalization and Data Science group (AG Data Science)
Location	Bundesallee 68, 38116 Braunschweig, Germany
Duration	16 March 2026 to 15 June 2026 (3 months)
Supervisor	Dr. Florian Schneider, Head of the Regionalization and Data Science group
Study programme	M.Sc. Organic Agriculture and Food Systems, University of Hohenheim
Main topic	Afforestation and Deforestation Effects on European Topsoil Carbon
EU project	AI4SoilHealth: Accelerating Collection and Use of Soil Health Information Using AI Technology to Support the Soil Deal for Europe and the EU Soil Observatory (Grant Agreement No. 101086179)

Main tools and data

R (tidyverse, terra, sf); QGIS; German Agricultural Soil Inventory (BZE-LW); LUCAS-SOIL data; digitised, georeferenced aerial photographs (1950 to present)

2. The Host Institution

2.1 The Johann Heinrich von Thünen Institute

The Johann Heinrich von Thünen Institute, usually called the Thünen Institute, is the German Federal Research Institute for Rural Areas, Forestry and Fisheries. As a federal research institute, it is an unincorporated institution under public law and an independent higher federal authority within the portfolio of the German Federal Ministry of Agriculture, Food and Regional Identity (BMLEH). Its head office is at Bundesallee 50 in Braunschweig, and the institute is led by its President, Prof. Dr. Birgit Kleinschmit. It takes its name from Johann Heinrich von Thünen (1783 to 1850), an agricultural economist whose ideas on land use and spatial economics are still widely cited today.



Figure 1. The Thünen Institute in Braunschweig. *The site in Braunschweig houses the central administration and a large part of the agricultural research (Photo: Thünen-Institut/Nadine Zirbes).*

The Thünen Institute consists of 15 specialised institutes with socioeconomic, ecological and technological expertise. Together they carry out research and provide policy advice on rural areas, agriculture, forests and fisheries. Much of the agricultural research is based in Braunschweig, while other institutes are located elsewhere in Germany. Because the institute combines research with direct advice to the federal government, its work often feeds straight into national policy and reporting, for example on soils and greenhouse gas emissions.



Figure 2. Bust of Johann Heinrich von Thünen (1783 to 1850). *The institute is named after the agricultural economist whose work on land use and spatial economics is still cited today.*

2.2 The Institute of Climate-Smart Agriculture

I did my internship at the Thünen Institute of Climate-Smart Agriculture (Institut für Agrarklimaschutz), one of the 15 specialised institutes. It carries out research and provides policy advice on climate change mitigation measures in agriculture. Its central questions are how land use and management affect the exchange of carbon dioxide, nitrous oxide and methane between the soil and the atmosphere, and how farming practices can be adapted to protect soil carbon and reduce emissions while keeping the land productive.

The institute is currently running the first iteration of the German Agricultural Soil Inventory (Bodenzustandserhebung Landwirtschaft, BZE-LW). This is a systematic survey of agricultural soils across the whole country, and it supplies the reference data that German soil carbon reporting relies on. Alongside this national monitoring, the institute uses European datasets, laboratory analysis, spatial modelling and data science to estimate soil organic carbon stocks, understand what controls them and follow how they change over time. Within the institute I was part of the Regionalization and Data Science group, led by my supervisor Dr. Florian Schneider. The group works on the statistical and computational analysis of large soil carbon datasets.

2.3 The Regionalization and Data Science Group and My Workplace

The Regionalization and Data Science group (AG Data Science) brings together soil science, statistics and programming. Its members work on large datasets such as the soil inventory, build models of soil carbon stocks and their changes, and produce the analyses that feed into national reporting and scientific publications. One example of the group's work was hanging right next to my desk: a poster on land use change and soil organic carbon in German agriculture by David Emde, Axel Don and Florian Schneider. It deals with how quickly soil carbon changes when grassland is turned into cropland and the other way round, which is closely related to my own project, and it was a useful reference whenever I thought about how to present my results.

The group also contributes to the EU project AI4SoilHealth, "Accelerating Collection and Use of Soil Health Information Using AI Technology to Support the Soil Deal for Europe and the EU Soil

Observatory” (Grant Agreement No. 101086179). Through this I was able to see how the work of a national institute connects with soil research and policy at European level (see Section 6.4).



Figure 3. My workplace in the Regionalization and Data Science group. *I did all of my analysis at this desk. The poster on the right shows the group’s research on land use change and soil organic carbon in German agriculture.*

For an internship that brought together soil science and data analysis, this was a very good place to be. My work was not a single lab or field task. Instead, it was built around research questions on soil organic carbon, the datasets that could answer them, and the steps needed to get from raw measurements to results that actually mean something. The rest of this report covers the scientific background, the two datasets I used, the research project I carried out and the results I obtained.

3. Scientific Background: Soil Organic Carbon

3.1 What Soil Organic Carbon Is

Soil organic carbon (SOC) is the carbon stored in soil organic matter, which is made up of the decomposed and partly decomposed remains of plants, roots, microorganisms and animals, along with the substances they turn into in the soil. It is the largest carbon store in the terrestrial biosphere and holds more carbon than the atmosphere and all living vegetation put together. Since this store is so big, even a small relative change in soil carbon means a large absolute exchange of carbon dioxide with the atmosphere. That is why soil organic carbon is central to both soil fertility research and climate policy, and why it was the focus of my internship at an institute working on climate smart agriculture.

Most carbon gets into the soil through plants. During photosynthesis, plants take carbon dioxide from the air and build it into their tissue. This carbon then reaches the soil as leaf and stem litter on the surface, as roots die and are replaced, and as organic compounds released by living roots into the surrounding soil. Carbon leaves the soil mostly through decomposition. Soil microorganisms feed on organic matter for energy and breathe much of it back out as carbon dioxide. Whether a soil gains or loses carbon over time therefore comes down to the balance between these two flows: carbon coming

in from plants and carbon going out through microbial respiration. Nearly every land use practice, soil property and climate factor that affects soil carbon does so by tipping this balance one way or the other.

3.2 What Controls How Much Carbon a Soil Holds

Several factors decide how much carbon a soil can hold at a given site, and they are important for understanding the results later in this report. The main ones are climate, land use and management, soil texture and soil type.

- **Climate.** Temperature and moisture have the strongest overall effect on soil carbon. Warmth speeds up microbial decomposition, so warmer soils usually lose carbon faster and store less of it. Cool conditions slow decomposition down and let carbon build up. Waterlogged soils with little oxygen slow decomposition even more, which is why the wettest and coldest soils are often the richest in carbon.
- **Land use and management.** Whether land is used as forest, grassland or cropland has a big effect on how much carbon goes in and how much is lost. Undisturbed grassland and forest keep adding carbon through dense root systems and litter. Cultivation and ploughing, on the other hand, break up the soil, expose protected organic matter to microbes and speed up carbon loss.
- **Soil texture.** The share of fine clay particles affects how much carbon a soil can stabilise. Clay has a huge surface area that organic molecules attach to, and this protects them from being broken down by microbes. Soils with more clay therefore usually hold and protect more carbon than sandy soils.
- **Soil type.** Organic and peat soils form under permanently wet conditions and can store huge amounts of carbon that have built up over thousands of years. They are fairly rare, but they account for the highest carbon stocks. They are also very vulnerable. Once they are drained, oxygen reaches the stored carbon and it can be released quickly.

These factors do not work separately but interact with each other. A cool, wet site under permanent forest on clay soil will hold far more carbon than a warm, dry, ploughed field on sandy soil. A big part of my analytical work was trying to separate the effects of these factors, and especially to tell the effect of land use and land use change apart from the effect of climate. The two datasets described in the next sections allowed me to do this both for Germany and for a wider area of Europe.

3.3 Soil Monitoring in Germany and Europe

Reliable statements about soil carbon depend on good monitoring data. In Germany several soil monitoring programmes run side by side, operated by the federal states, the German Weather Service, the Federal Environment Agency, the Thünen Institute and, at European level, the Joint Research Centre. In terms of sampling points, the two Thünen inventories are by far the largest: the agricultural soil inventory (BZE Agriculture) with 3,104 sites and the forest soil inventory (BZE Forest) with 1,873 sites. The European LUCAS survey adds 2,845 sampling points in Germany alone.



Figure 4. Soil monitoring programmes in Germany. Presentation by Axel Don at the AI4SoilHealth Consortium Meeting (24.03.2026). The boxes show the number of sampling points of each programme; BZE Agriculture (3,104) and LUCAS (2,845) are the two datasets I used.

Seeing this overview early in my internship helped me understand why the two datasets I worked with are so valuable. The BZE gives a very detailed, deep picture of German agricultural soils down to one metre, while LUCAS samples topsoil in the same way across all EU member states. Using both allowed me to look at soil carbon first at national and then at European scale.

4. Objectives of the Internship

At the start of the internship I agreed on the following learning objectives with my supervisor:

- To understand how large soil monitoring datasets are built, structured and used for soil carbon reporting.
- To learn the full analytical workflow in R, from data cleaning and joining tables to statistics and figures ready for publication.
- To work out how land use, climate and soil properties together shape soil organic carbon.
- To carry out an independent research project on the effect of afforestation and deforestation on topsoil carbon in Europe.
- To learn how to reconstruct land use history from historical aerial images and document it in a reproducible way.
- To learn to collect and process spatial data in a geographic information system (QGIS).
- To contribute to an EU research project and gain experience in organising an international project meeting.
- To experience how a federal research institute works, and how its research connects to policy questions on land use and climate.

These objectives guided my work throughout the three months and are the basis for the reflection later in this report.

5. Internship Timeline

The internship followed the natural order of a research project: getting to know the data, exploring it, building the main dataset, analysing it and writing up the results. The table below gives an approximate overview of how my work was distributed over the three months.

Period	Main focus
16 to 31 March 2026	Introduction to the institute and the Regionalization and Data Science group; planning, logistics and onsite support for the AI4SoilHealth Consortium Meeting in Braunschweig (23 to 25 March); setting up R and QGIS; reading on soil organic carbon
April 2026	Exploratory analysis of the German Agricultural Soil Inventory: data structure, carbon stock and concentration, depth profiles, land use, peat, clay, bulk density and pH
May 2026	Planning the main project; selecting LUCAS-SOIL sites in Spain, France, Baden-Württemberg and Sweden; systematic examination of digitised, georeferenced aerial photographs (1950 to present) in QGIS for about 1,000 sites; integrating the land use time series with published soil carbon measurements
June 2026 (until 15 June)	Classification of land use transitions; reproducible analysis pipeline in R; publication ready figures, tables and descriptive statistics; interpretation with my supervisor; writing up and presenting the results

Some tasks, especially reading literature and improving figures, did not belong to one month only but continued throughout the internship. The table therefore shows when each activity was the main focus, not a strict schedule.

6. Tasks and Project Work

The internship was one connected research project on soil organic carbon, done in two phases. In the first phase I explored the German Agricultural Soil Inventory (BZE). This helped me get used to soil carbon data, learn how carbon is measured and reported, and practise the statistics and plotting workflow used at the institute. The second and main phase was my own research project on how afforestation and deforestation affect topsoil organic carbon across Europe. For this I used LUCAS soil data combined with a reconstruction of the land use history at each site. The BZE work gave me the background and the technical skills, while the European study on afforestation and deforestation was the main scientific output of the internship and makes up the core of this report.

Alongside this research, I contributed to the EU project AI4SoilHealth, most of all by helping to organise its consortium meeting in Braunschweig, which is described in Section 6.4.

6.1 Working Environment and Analytical Workflow

I used the same tools in both phases. All data cleaning, statistical analysis and plotting were done in the open source programming language R. I used the tidyverse packages for data handling and plotting, and the packages terra and sf for raster and vector spatial data. Spatial data collection and

processing, in particular the work with historical aerial photographs, was done in the open source geographic information system QGIS. This meant handling large soil data tables, joining them through shared site identifiers, calculating summary statistics and producing the figures shown in this report. Because the workflow stayed the same, what I learned in the first phase carried straight over to the main project.

Building reproducible analysis pipelines from the first week also meant learning good habits: keeping raw data untouched, writing scripts that can be run again from the start, and saving every figure from code instead of editing it by hand. These habits made it much easier to update results whenever my supervisor suggested a change.

6.2 Exploratory Analysis of the German Agricultural Soil Inventory

The German Agricultural Soil Inventory is a systematic survey of Germany's agricultural soils, run by the Thünen Institute. The version I used contains 3,104 sampling sites across Germany, and each site was sampled in the same standard way down to one metre. The data are openly available through the Open Agrar repository. The inventory was a good way to start working with soil carbon data. It is large, well organised and covers the whole country, and it shows the main controls on soil carbon described above very clearly.

The inventory consists of three linked tables. The site table has 3,104 rows with the location, coordinates, land use and carbon stocks of each sampling point. The horizon table has 15,219 rows describing the soil layers identified in the field. The laboratory table has 17,189 rows of measurements taken at five standard depths: 0 to 10 cm, 10 to 30 cm, 30 to 50 cm, 50 to 70 cm and 70 to 100 cm. All three tables share a unique site identifier (PointID), so measurements, field descriptions and site information can be combined for analysis. The sites belong to three land use classes: cropland (2,234 sites), grassland (820 sites) and special crops such as vineyards and orchards (50 sites).

The first thing the inventory made clear to me was the difference between two ways of measuring soil carbon. Carbon concentration is the amount of carbon per unit mass of soil, given in grams of carbon per kilogram of soil. It tells you how rich in carbon the soil material is. Carbon stock is the total amount of carbon stored under a certain area down to a certain depth, given in megagrams of carbon per hectare. It tells you how much carbon is actually in the ground. The two are linked through bulk density and depth, so roughly speaking the stock is the concentration multiplied by bulk density and depth. This means two soils with the same concentration can have very different stocks if one is denser than the other. Both measures are skewed to the right across the inventory. Most sites have moderate values, but a few organic and peat soils have extremely high carbon and form a long tail in the distribution.

Another clear pattern is where in the profile the carbon sits. At most sites, the bulk of the carbon is close to the surface. When I calculated the share of total profile carbon found in the top 30 cm for each site, most sites were well above 50%, with a peak at around two thirds. So at a typical site, about two thirds of the carbon in the upper metre is in the top 30 cm. This makes the topsoil very important, since any disturbance of the upper 30 cm affects most of the stored carbon. Only a few sites, with deep carbon from old organic deposits, have most of their carbon further down.

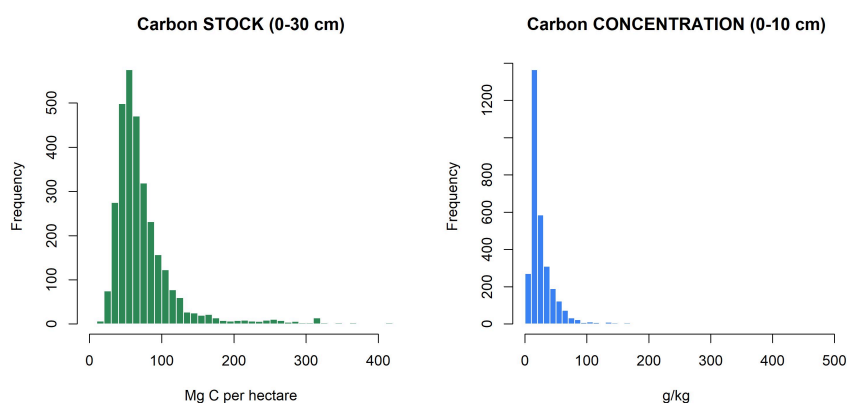


Figure 5. Two ways of measuring carbon across the German inventory. Left: carbon stock in the upper 0 to 30 cm (Mg C/ha). Right: carbon concentration (g/kg). Both distributions are skewed to the right, with a long tail of sites rich in organic matter.

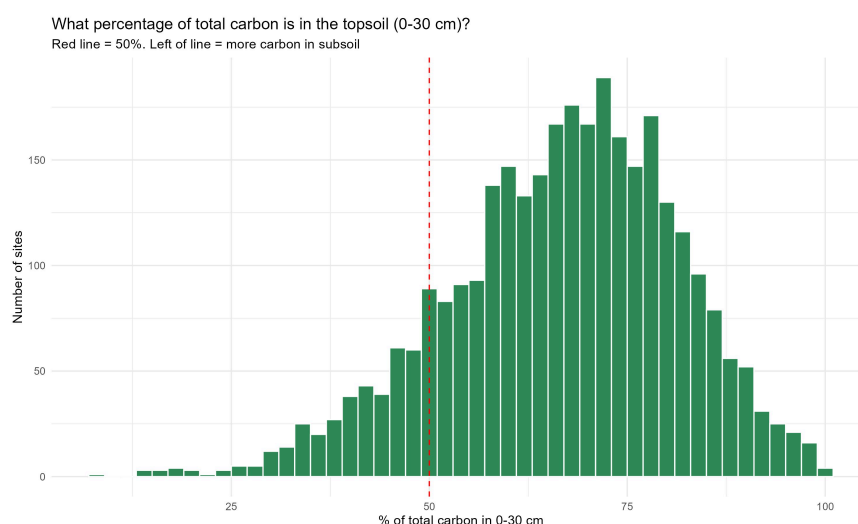


Figure 6. Most carbon is in the topsoil. Distribution of the share of total profile carbon held in the top 30 cm. Most sites lie to the right of the 50% line, with a peak near two thirds.

Splitting this pattern by land use shows that all three land use types store more carbon in the topsoil than in the subsoil, though the spread differs. Cropland and grassland have similar median topsoil shares of about 68 to 70%, but cropland varies more. Special crops have a slightly lower median of about 62%. The more even spread of carbon with depth under cropland comes from ploughing, which mixes the upper soil and moves carbon downward through the cultivated layer.

The drop in carbon with depth becomes obvious when mean carbon concentration is plotted for the five standard depths and each land use. In all three land uses, carbon is highest in the top 10 cm and falls sharply further down. Grassland starts at about 45 g/kg at the surface, almost two and a half times the cropland value of about 18 g/kg. The gap between them gets smaller with depth but never disappears, and grassland has more carbon at every depth. Below about 50 cm the land uses come together at roughly 5 g/kg, which shows that management mainly affects the surface and its influence fades deeper down. The distance between the grassland and cropland curves is basically a direct measure of how much carbon cultivation costs.

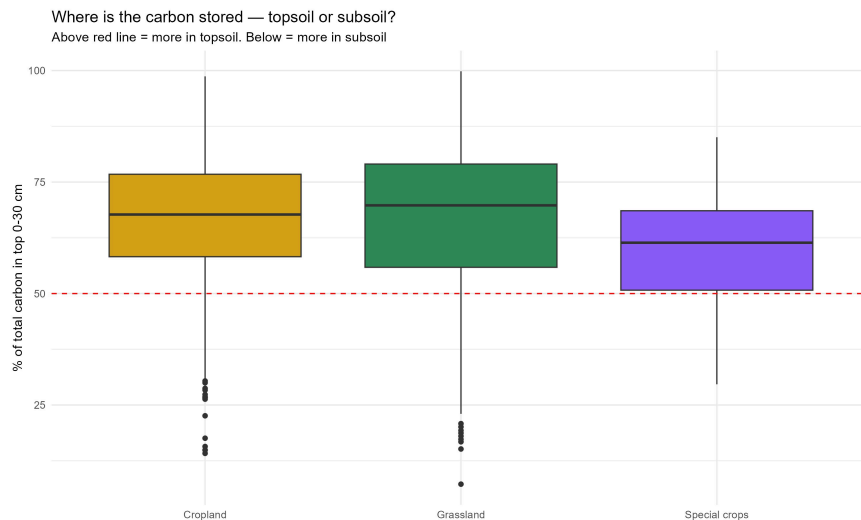


Figure 7. Topsoil versus subsoil carbon by land use. All land uses store most of their carbon in the topsoil. Cropland shows a wider spread, which fits with ploughing mixing the profile.

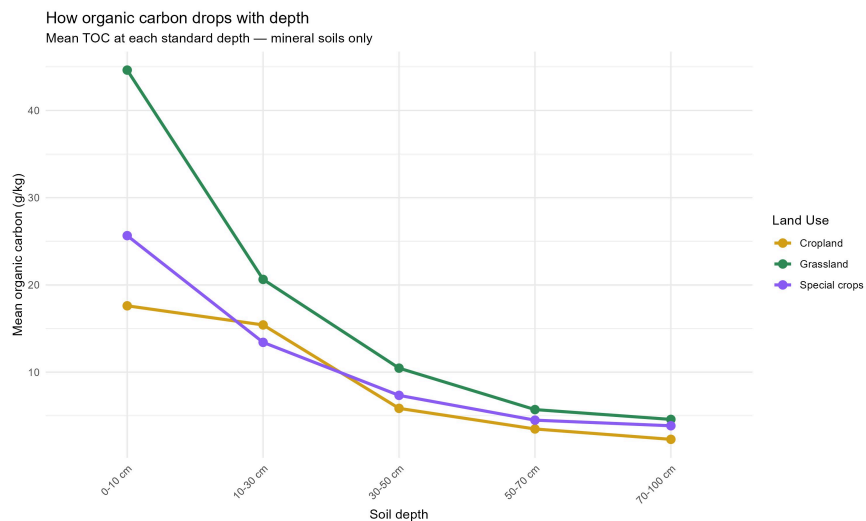


Figure 8. Carbon declines sharply with depth and grassland stays ahead. Mean organic carbon concentration at each standard depth for cropland, grassland and special crops (mineral soils). Grassland is highest at every depth.

Carbon storage also differs a lot between the German federal states. Comparing mean carbon stock by state, the coastal lowland states of Schleswig-Holstein and Niedersachsen come first with about 95 Mg C/ha, because of their wet lowlands and peatlands. Sachsen, which is drier and mostly cropland, is lowest at about 55 Mg C/ha. A range of 55 to 95 Mg C/ha shows how strongly geography shapes carbon storage. The obvious next question was why these regional differences exist.

Part of the answer is the mix of land uses in each state. States with high carbon stocks usually have a large share of grassland, while the low carbon states are mostly cropland. Some smaller states stand out. Saarland has a notable share of special crops because of its wine region, and the city state of Bremen is almost all grassland. So the land use mix explains a lot of the difference between states, but not all of it, since wetlands, soil type and climate also play a role. Putting plots side by side like this, where one figure raises a question and the next one partly answers it, taught me a lot about how to tell a story with data.

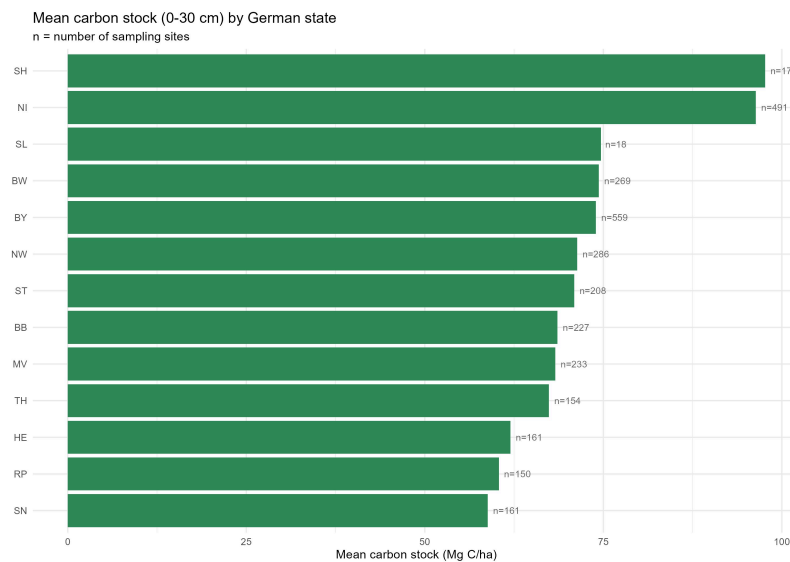


Figure 9. Carbon stock varies by federal state. Mean topsoil carbon stock (Mg C/ha) by German state, with sample sizes shown. Coastal lowland states lead and drier cropland regions are lowest.

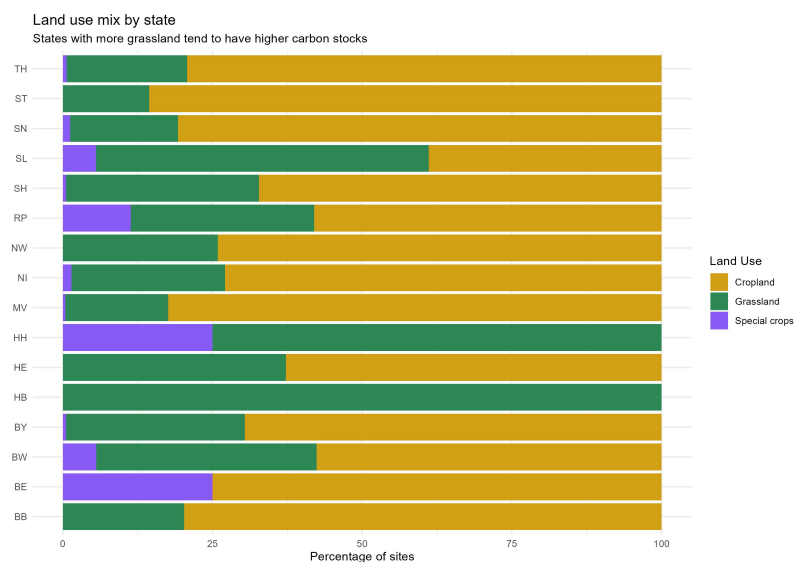


Figure 10. The land use mix helps explain the differences between states. Share of cropland, grassland and special crops in each state. High carbon states have more grassland, while low carbon states are mostly cropland.

The single biggest control on carbon in the inventory is peat. When the sites are split into mineral and peat soils, the median carbon stock of peat soils is about 240 Mg C/ha, around four times that of mineral soils at about 60 Mg C/ha. Only 146 of the 3,104 sites, fewer than 5%, are peat soils. Even so, they make up the high end of carbon storage and cause the long tail seen in the distributions. Peat is thousands of years of dead plant material that has been preserved because the soil is waterlogged and has no oxygen. If these soils are drained for farming, oxygen gets in, microbes become active again and the old carbon is released as carbon dioxide. For this reason, protecting peatlands is one of the most cost effective climate measures available in agriculture.

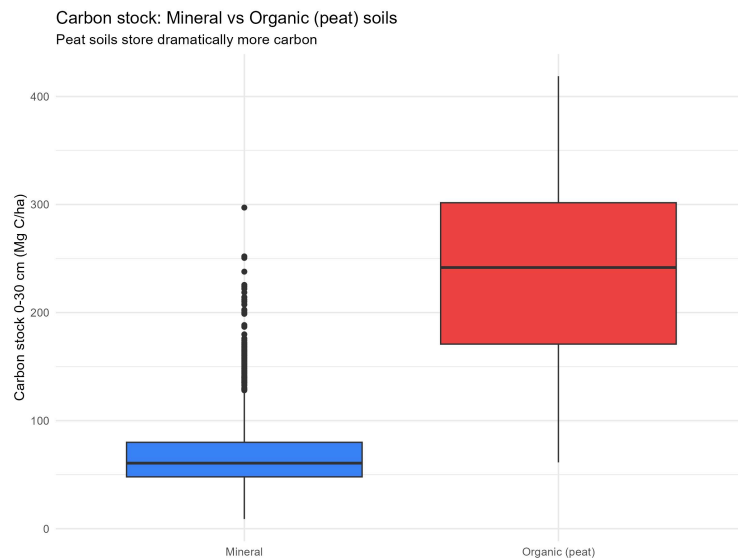


Figure 11. Peat soils are concentrated carbon stores. Carbon stock of peat versus mineral soils. Peat sites make up fewer than 5% of the inventory but hold about four times the median carbon.

Looking at carbon stock directly by land use confirms what the rest of the inventory shows. The median grassland stock of about 88 Mg C/ha is clearly higher than the cropland median of about 58 Mg C/ha. The grassland distribution is also taller, meaning it varies more, because it includes some sites rich in organic matter. Cropland has several high outliers, which are most likely peat soils that are being farmed. Once again, the difference between cropland and grassland reflects the carbon lost through ploughing.

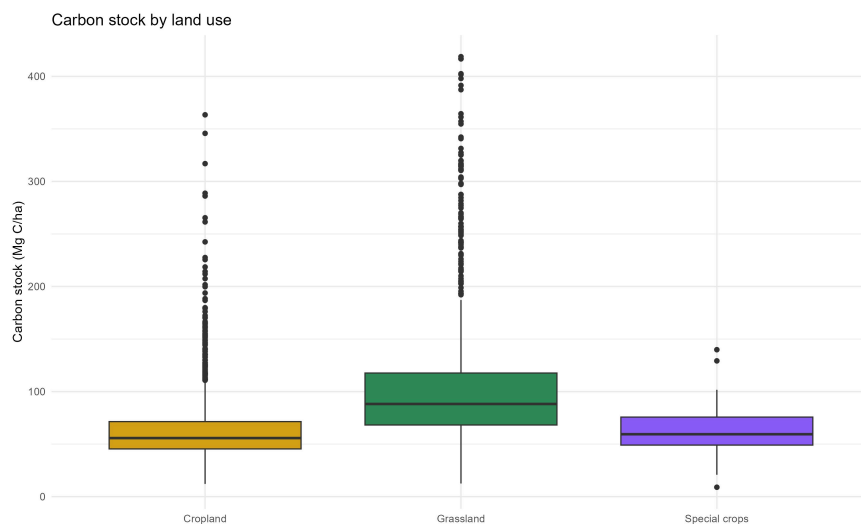


Figure 12. Carbon stock by land use. Topsoil carbon stock for cropland, grassland and special crops. Grassland holds clearly more carbon than cropland.

Two relationships based on soil physics complete the exploratory analysis. The first is about soil texture. When organic carbon is plotted against clay content, there is an upward trend, meaning that sites with more clay tend to hold more carbon. Clay particles are tiny and have a very large surface area. Carbon molecules bind to this surface and are protected from microbes. Sandy soils with little clay cannot protect carbon in the same way and lose it faster. At the same clay content, grassland sites lie above cropland sites, which shows that land use has an effect of its own, separate from texture.

The points are widely scattered, so clay is only one of several controls alongside climate, management and soil biology. Still, the relationship shows nicely how a natural soil property that cannot really be changed and a human decision about land use work together to set carbon storage.

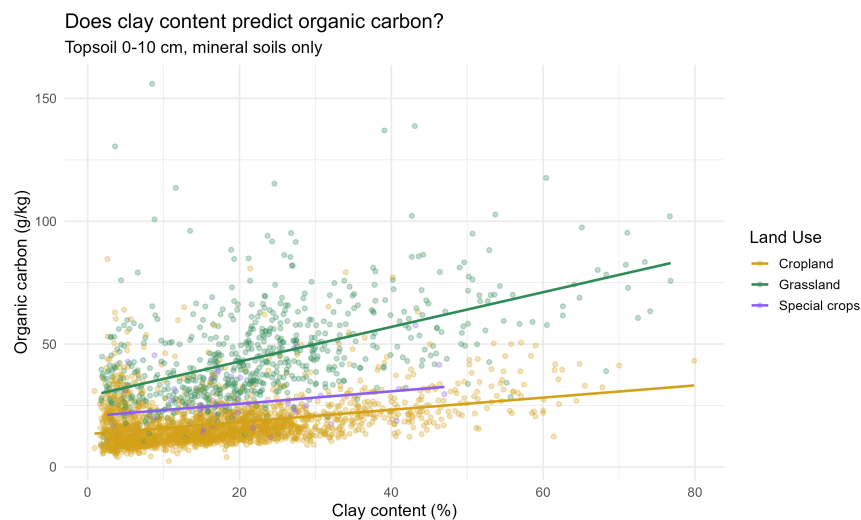


Figure 13. More clay is associated with more carbon. Organic carbon against clay content, coloured by land use. The upward trend reflects carbon being protected on clay surfaces. Grassland lies above cropland at the same clay content.

The second relationship is with bulk density, which is the mass of soil per unit volume. Here the trend clearly goes down: the higher the bulk density, the lower the organic carbon. Organic matter is lighter than mineral particles, so a soil with a lot of carbon is lighter and more porous. Carbon also helps create pore space and structure, which lowers the density even further. Grassland soils group at low density and high carbon, and cropland soils at higher density and lower carbon. This was the tightest relationship I looked at, because bulk density and carbon content are physically linked.

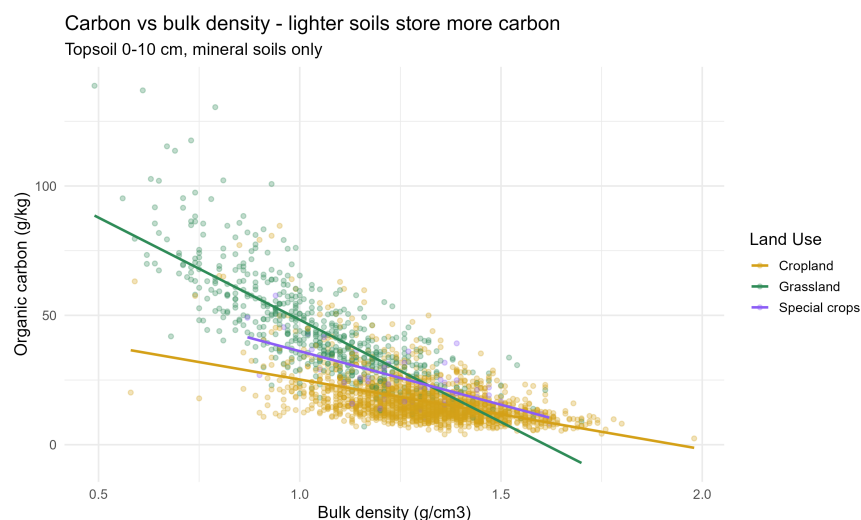


Figure 14. Lighter soils hold more carbon. Organic carbon against bulk density. The strong negative relationship reflects the direct physical link between organic matter content and soil density.

Finally, soil pH by depth and land use gives a clear fingerprint of management. Cropland has a higher pH than grassland at every depth. This is because farmers lime cropland to correct acidity and help crops grow, while grassland gets no lime and keeps acidifying naturally. pH rises slightly with depth in

all land uses because of the less acidic parent material underneath. Special crops have the highest pH, which fits with the intensive management of vineyard and orchard soils. In short, the pH pattern directly reflects how the land is farmed.

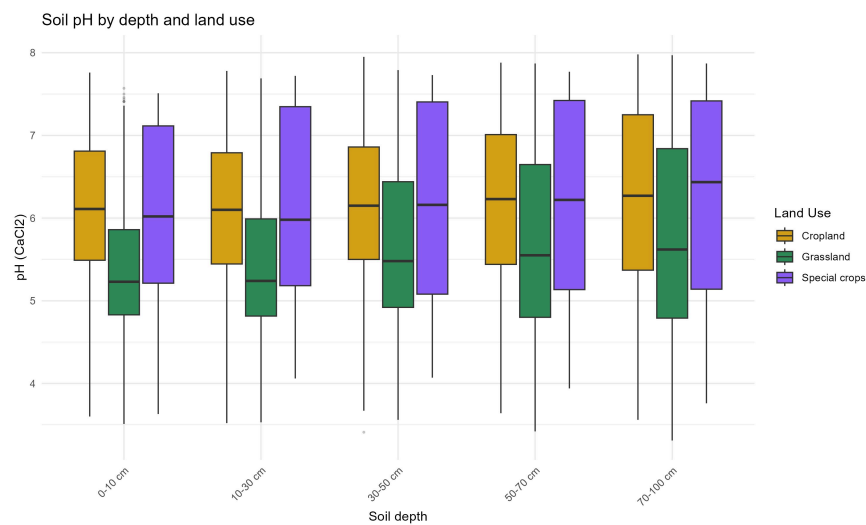


Figure 15. Soil pH by depth and land use. Cropland is consistently less acidic than grassland because of liming, and special crops show the highest pH. pH rises slightly with depth.

Overall, the exploratory analysis of the German Agricultural Soil Inventory led to several conclusions that set up the main project. How much carbon a soil stores depends on soil type, texture, structure and land use together. Grassland stores much more carbon than cropland, because undisturbed roots keep feeding carbon into the soil while ploughing breaks it down. Peat soils are rare but cannot be replaced as carbon stores. And while soil type and texture are fixed, land use is something people can change. This last point, that land use is the part we can actually control, led directly to the main question of my project: what happens to soil carbon when land use really does change, and especially when land is turned into forest or forest is cleared?

6.3 Main Project: Afforestation and Deforestation Effects on European Topsoil Carbon

The main project looked at how turning land into forest (afforestation) or clearing forest (deforestation) affects the organic carbon content of mineral topsoil across Europe. It also looked at how this effect depends on the time since the change and on climate. Soil organic carbon is the largest carbon store on land, and changing land to or from forest is expected to change how much carbon the topsoil holds. The problem is that this effect is slow, takes decades to show, and is mixed up with the strong effect of climate on soil carbon. Separating the land use signal from the climate signal was the main scientific challenge of the project.

6.3.1 Research questions

The project was guided by three specific questions:

1. After afforestation or deforestation, how do long term soil organic carbon levels in mineral topsoil compare with those under land use that has stayed the same?
2. How does soil organic carbon change with the time that has passed since the land use change?

3. How does climate, in particular mean annual temperature and mean annual precipitation, affect this relationship?

To answer these questions under a wide range of conditions, I worked with four European regions that together cover a long climate gradient from south to north: Spain and France in the warmer and drier south and west, Baden-Württemberg in southwestern Germany, and Sweden in the cool, wet north. These regions were chosen on purpose because their temperatures and rainfall are so different, and this is exactly what makes it possible to separate the effect of land use from the effect of climate. The wider research proposal included more countries, but these four were used for the analysis because they give a strong climate gradient and have historical images available for reconstructing land use history.

6.3.2 Data and method

The analysis brought together two datasets, linked at the level of individual sampling sites. The first was the LUCAS soil dataset. LUCAS (the Land Use and Coverage Area frame Survey) is a harmonised soil survey run by the European Commission, in which topsoil is sampled and analysed in the same way in all member states over repeated survey rounds. From the published LUCAS-SOIL data I took, for each site in the four study regions, the mineral topsoil organic carbon concentration, its recent trend between 2009 and 2018, and the site climate as mean annual temperature and mean annual precipitation.

The second dataset was a reconstruction of land use history that I put together myself. The carbon effect of afforestation or deforestation depends on how long ago the change took place, but the land cover change records in LUCAS only go from about 2006 to 2018. That is too short to pick up changes that happened decades earlier. To get around this, I collected historic land use data for about 1,000 LUCAS-SOIL sites by systematically examining digitised, georeferenced aerial photographs from 1950 to the present. I did this in QGIS, where I viewed the images for each site, recorded the land cover at three or more points in time and processed the resulting spatial data. This captures the real long term land use change at each site, which LUCAS on its own misses, and it was the main methodological contribution of the project.

I then integrated the historic land use time series with the published soil organic carbon measurements of the same LUCAS-SOIL sites, using their shared site identifier. This created a unified dataset ready for statistical analysis. Based on the reconstruction, I classified every site by its type of transition: persistent forest, persistent cropland or persistent grassland; conversion to forest (afforestation); conversion from forest (deforestation); or conversion between cropland and grassland. I also noted the approximate year of each transition. I only used mineral topsoil and left out strongly organic soils, and I kept only sites with either unchanged land use or one clear change, so that each site had a clear land use history. After these restrictions, about 900 sites remained for the analysis. A separate metadata table recorded the year, land use and source of every reconstructed observation, for archiving and so that the work can be reproduced. All merging, classification and analysis were built as a reproducible pipeline in R.

Combining a harmonised European soil dataset with a land use reconstruction built for this purpose, across a deliberately chosen climate gradient, made it possible to compare soil carbon between sites

that had changed and sites that had not, while taking the strong background effect of climate into account. The results of this comparison are presented in the next section.

6.3.3 Results

The results follow the order in which I built up the analysis. I start with the study area and the data, then look at the land use changes found in the reconstruction, the link between land use and soil carbon, the effect of climate and, finally, the recent carbon trend. Every figure is one of the publication ready graphics I produced in R during the internship, together with the tables and descriptive statistics behind them.

The study area and the data

The basis of the analysis is how the sites are spread along the climate gradient. The map of the study area shows every LUCAS site, coloured by its current land cover. The four regions are clearly separate, from Spain and France in the south and west, through Baden-Württemberg, up to Sweden in the north. All three land cover types (cropland, grassland and woodland) appear along the whole gradient. Having a wide climate range with every land cover type present is what makes it possible to separate the land use effect from the climate effect.

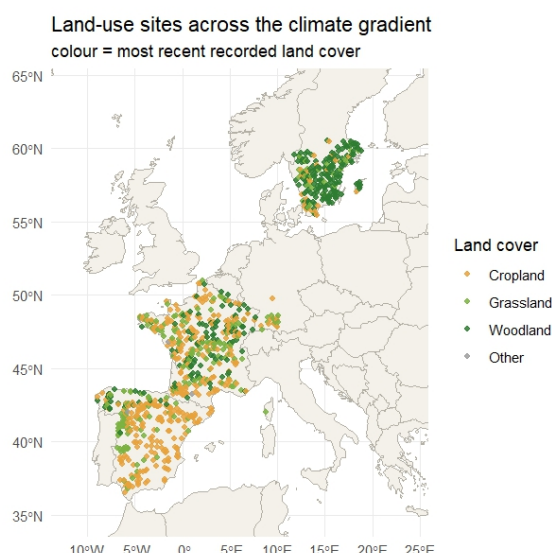


Figure 16. The study area. Each point is a LUCAS site, coloured by its current land cover, across the four regions along a climate gradient from south to north.

The land cover mix is different in each country. Sweden is mostly forest, Spain and France have more cropland and grassland, and Baden-Württemberg adds a smaller mixed group of sites. This matters because it affects where each type of land use change can happen. For example, a region with little forest offers few chances to observe deforestation. So each country adds its own land cover mix to the gradient.

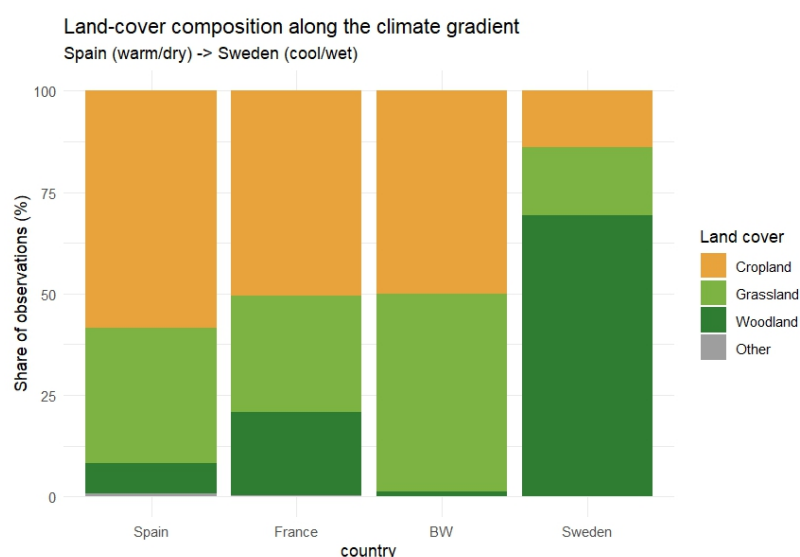


Figure 17. Land cover by country. Sweden is mostly forest, Spain and France are more agricultural, and Baden-Württemberg is a smaller mixed set. Each region adds its own mix.

Land use change since the 1950s

The reconstruction shows that most sites have kept the same land use since the 1950s, but a noticeable minority have changed. The changes I was most interested in are afforestation, where land has become woodland, and deforestation, where woodland has been cleared. There are also some conversions between cropland and grassland, mostly in the agricultural south. Having real land use change over several decades in the data is what the whole soil carbon comparison below is based on.

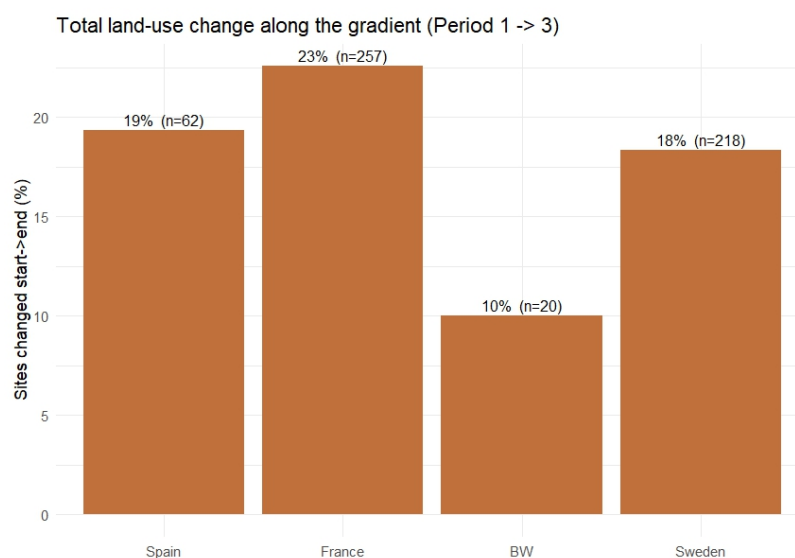


Figure 18. Land use change found in the reconstruction. Most sites are unchanged, but a noticeable minority changed. Afforestation and deforestation are the main transitions of interest, with some conversion between cropland and grassland in the south.

When the transitions are mapped, it becomes clear that land use change is regional and not evenly spread. Afforestation is concentrated in Sweden and northern Spain, while deforestation is rare everywhere in the dataset. Because change happens so unevenly, this has to be kept in mind when reading the carbon results.

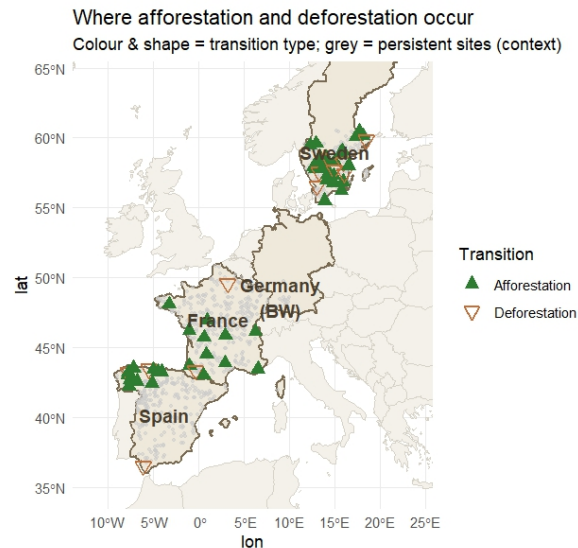


Figure 19. Where afforestation and deforestation occur. *Afforestation is concentrated in Sweden and northern Spain, and deforestation is uncommon in all regions. Land use change is regional rather than uniform.*

Conversions between cropland and grassland show the opposite pattern. They are concentrated in France and Spain, the agricultural core of the dataset, and hardly occur in Sweden. The kind of land use change therefore depends on the region: afforestation in the north and switches between farmland types in the south.

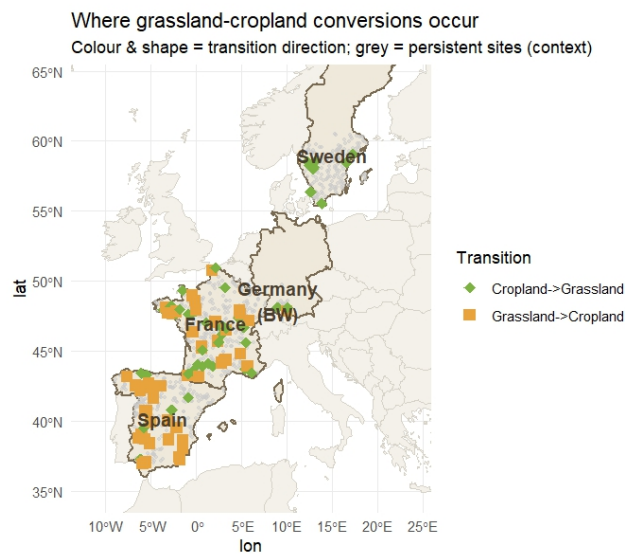


Figure 20. Where conversions between grassland and cropland occur. *These conversions are concentrated in France and Spain and hardly occur in Sweden, a pattern typical of southern and central Europe.*

Soil carbon and land use change

The key result is about how much carbon is held under each transition type. Sites that involve woodland, whether through afforestation or deforestation, hold about twice as much soil organic carbon as sites that have stayed cropland or grassland. The woodland signal dominates, so any land use history that includes forest leaves behind a soil rich in carbon. The comparison was based on 737

unchanged sites, 47 afforestation sites and 12 deforestation sites. Because there are so few deforestation sites, that group has to be treated with caution throughout.

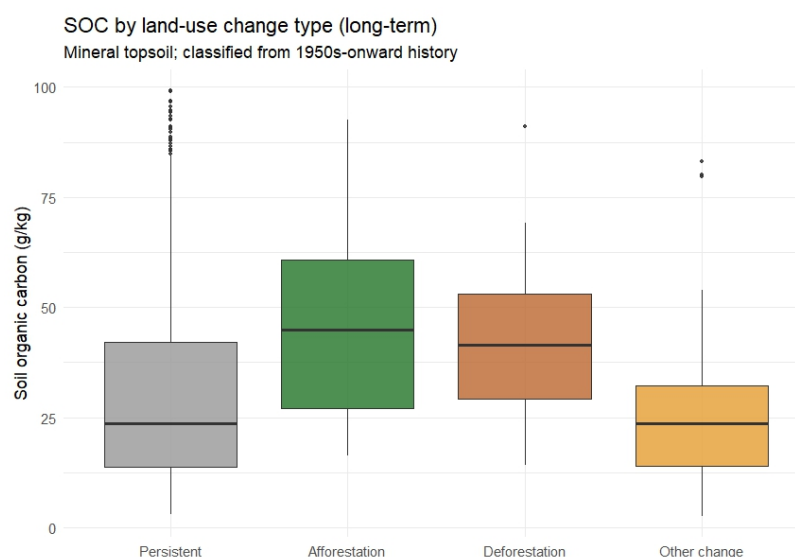


Figure 21. Soil carbon by type of long term land use change. *Afforestation and deforestation sites hold about twice the carbon of unchanged cropland or grassland sites. A history involving woodland leaves a lasting high carbon signature.*

For afforestation, I compared sites that had been turned into forest with land that stayed without forest, which is what they started as. The afforested sites are clearly higher, at about 45 g/kg compared with about 18 g/kg for land that never became forest. Put simply, several decades under forest build up much more topsoil carbon than staying as cropland or grassland. The afforested sites come from Sweden, Spain and France.

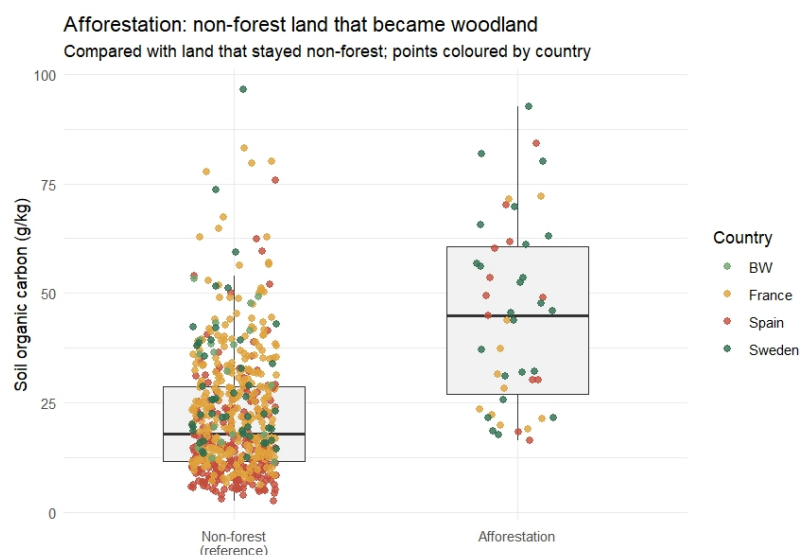


Figure 22. Afforested land holds more carbon than land without forest. *Afforestation sites (~45 g/kg) compared with land that stayed without forest (~18 g/kg). Turning land into forest goes along with clearly higher topsoil carbon.*

For deforestation, I compared cleared woodland sites with permanent woodland that was never cleared and with land that was never woodland. The deforested sites lie below permanent woodland but above land that was never forest. This suggests that clearing forest goes along with losing carbon

compared with keeping the forest, but that some of the old forest's legacy stays in the soil. With only 12 deforested sites, though, this result is a hint rather than a firm finding. It is best read as matching the afforestation result in reverse.

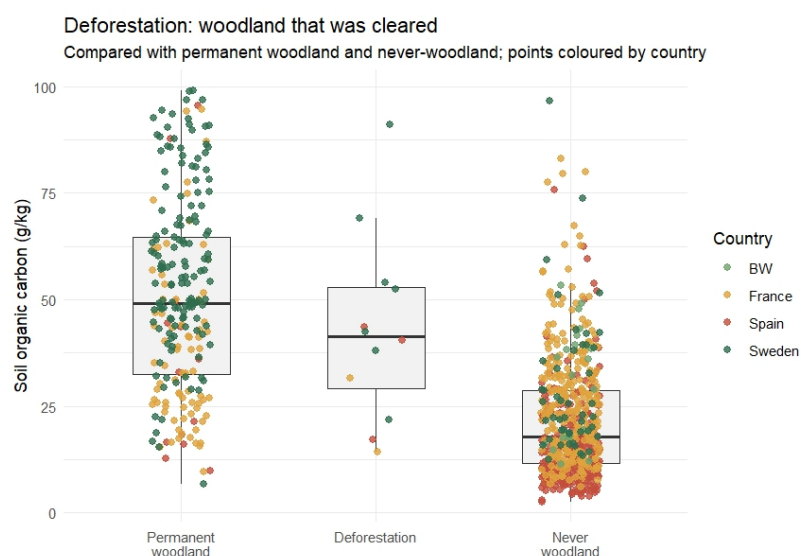


Figure 23. Cleared woodland lies between forest and land without forest. Deforestation sites compared with permanent woodland and land that was never woodland. Cleared sites are below permanent forest but above land that was never forest (n=12, indicative only).

The influence of climate

Climate turned out to be the strongest single control on soil carbon in the dataset, which is why separating it from the land use effect mattered so much. Plotting soil organic carbon against mean annual temperature shows a clear negative relationship, with a correlation of about -0.58 . Warmer sites hold much less carbon because the heat speeds up decomposition and stops carbon from building up. Precipitation has a weaker positive effect, with a correlation of about 0.24 . Climate therefore sets the baseline level of soil carbon: warm sites are poor in carbon and cool sites are rich in it.

The key test of the project was whether land cover still matters once climate is taken into account. To check this, I compared soil carbon by land cover within separate temperature bands. The same order holds in every band: woodland has the most carbon, cropland the least and grassland lies in between. At the same time, all values go down as temperature goes up, so the climate effect is still visible underneath. Since the land cover ranking stays the same in each temperature band, land cover affects soil carbon independently of climate. The two effects are real and can be separated, and this is the main methodological finding of the study.

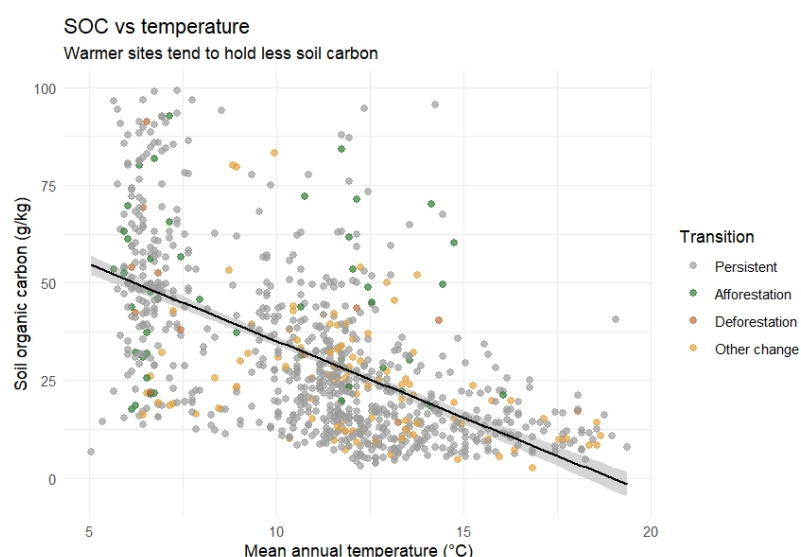


Figure 24. Temperature is the strongest control on carbon. Soil organic carbon against mean annual temperature ($r \approx -0.58$). Warmer sites hold far less carbon, and precipitation adds a weaker positive effect ($r \approx 0.24$).

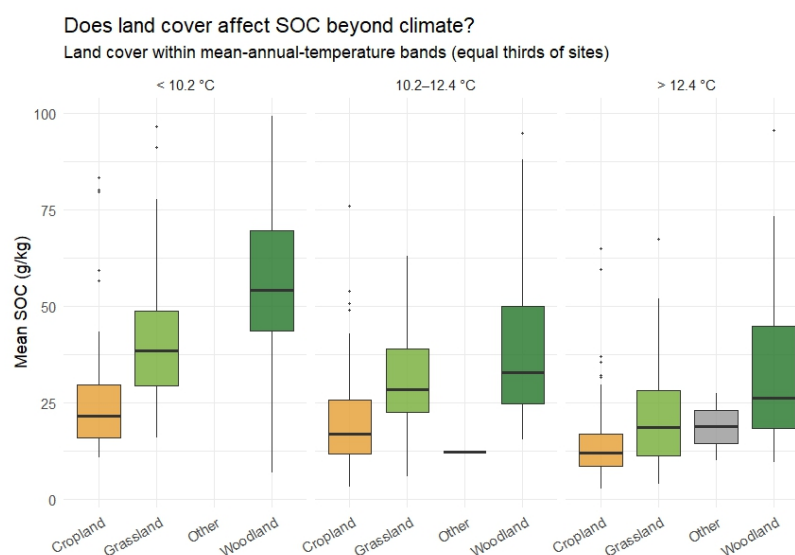


Figure 25. Land cover matters on top of climate. Soil carbon by land cover within three temperature bands. Woodland is highest and cropland lowest in every band, while all values fall as temperature rises.

The recent carbon trend

I also looked at the recent rate of change in soil carbon, based on the LUCAS time series from 2009 to 2018. Unrealistic rates were removed as sampling errors, and the width of each box shows how reliable each group is. When the trend is grouped by long term change category, all groups lie close to zero. This means a site's land use history does not clearly tell us whether its carbon is going up or down today.

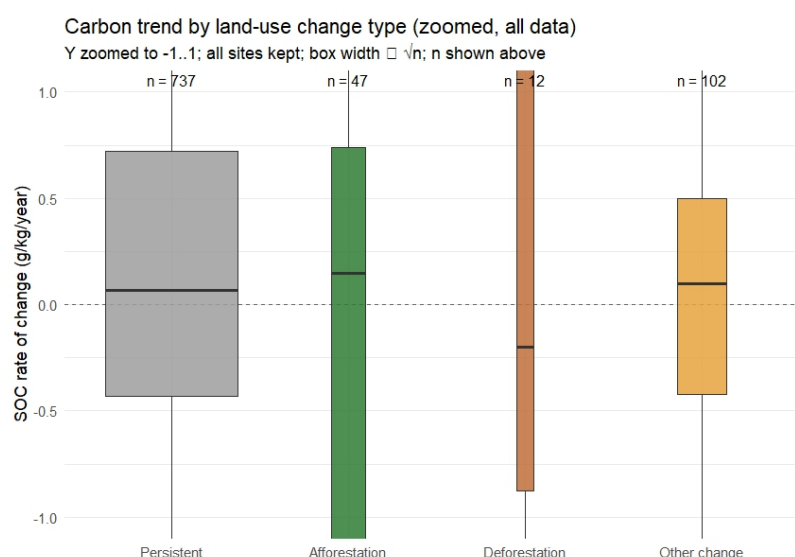


Figure 26. Recent carbon trend by long term change category. Rate of carbon change (2009 to 2018) grouped by historical change category. All groups lie close to zero, so past change does not predict the current trend.

When the same recent trend is grouped by what each site is today, a clearer pattern shows up. Sites that are woodland today lie above zero, meaning forest soils are still gaining carbon, while cropland and grassland stay roughly level. What the land is now predicts its current carbon trend better than how it has changed in the past.

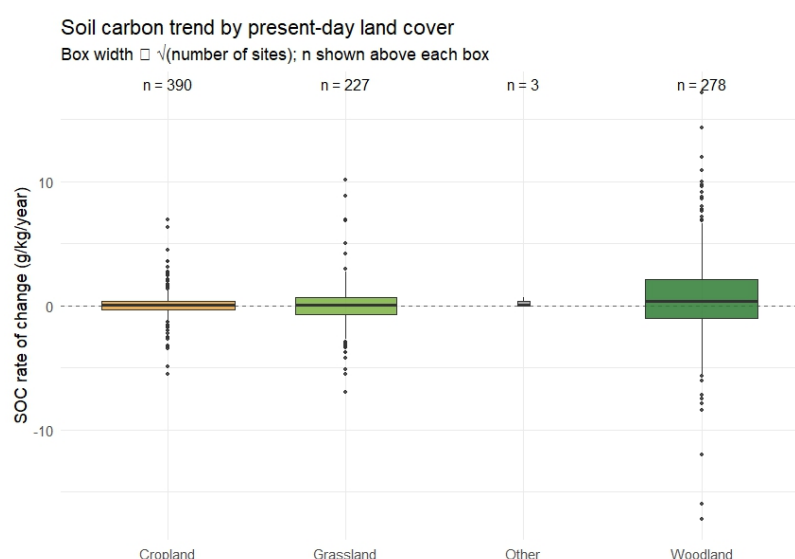


Figure 27. Today's woodland is still gaining carbon. Recent carbon trend grouped by current land cover. Woodland lies above zero, while cropland and grassland stay roughly level.

Lastly, today's level of soil carbon grouped by current land cover gives the clearest single result of the study. Woodland has a median of about 48 g/kg, grassland about 28 g/kg and cropland about 15 g/kg. Forest soils therefore hold about three times as much carbon as cropland. Current land cover is a strong and direct predictor of how much carbon a soil holds.

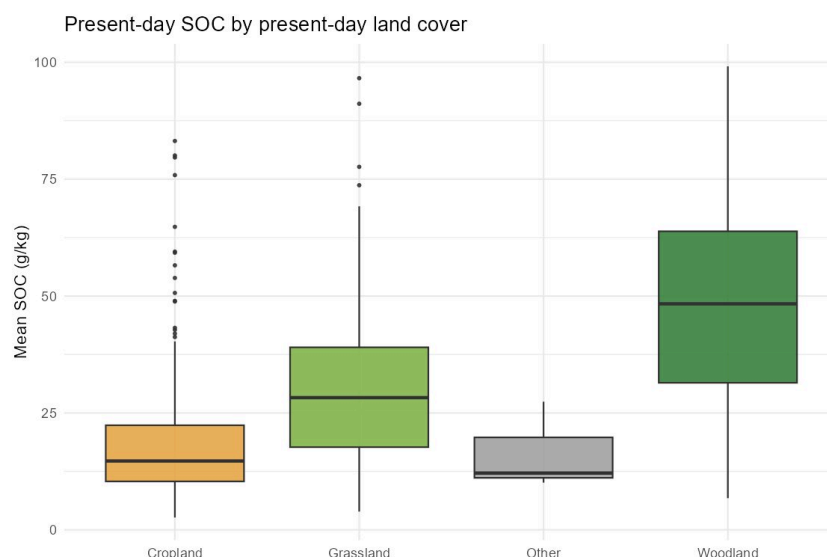


Figure 28. Current carbon by current land cover. Soil organic carbon by current land cover. Woodland (~48 g/kg) is far above grassland (~28 g/kg) and cropland (~15 g/kg), roughly a threefold difference.

6.3.4 Summary of findings

The project led to five main conclusions. First, woodland soils are rich in carbon. Sites with forest anywhere in their history hold about twice the carbon of cropland or grassland, and this forest signature lasts for decades. Second, afforestation builds up carbon. Afforested land holds far more carbon than the land without forest it came from, and deforestation shows the opposite, although the sample is small. Third, climate sets the baseline, and carbon falls sharply as temperature rises, but land cover still matters on its own within every climate band. Fourth, land use change is regional, with afforestation concentrated in Sweden and Spain and conversion between cropland and grassland limited to France and Spain. Fifth, the recent carbon trend follows current land cover rather than past change, and woodland is still gaining carbon today.

Caveats. There are some limitations to keep in mind. The deforestation group has only 12 sites, so its results are indicative rather than conclusive. The recent carbon trend only covers 2009 to 2018. Historical images for Spain basically stop around 2000, which makes the reconstruction less precise there. Finally, all the relationships found are correlations, not proof of cause. The analysis shows strong and consistent links between land use, climate and soil carbon, but on its own it cannot prove that land use change caused the differences in carbon.

6.4 Contribution to the EU Project AI4SoilHealth

Alongside my research, I contributed to the EU project AI4SoilHealth, “Accelerating Collection and Use of Soil Health Information Using AI Technology to Support the Soil Deal for Europe and the EU Soil Observatory” (Grant Agreement No. 101086179). The project works towards assessing the health of Europe’s soils. As the project banner shows, its main building blocks are novel on site soil analysis, a soil health data cube and a robust framework for selecting soil health indicators. Its results are meant to support the Soil Deal for Europe and the EU Soil Observatory.

Organising the AI4SoilHealth Consortium Meeting

My first major task, only one week after I joined the institute, was to assist with the planning, logistics and onsite support of the AI4SoilHealth Consortium Meeting, which took place in Braunschweig from 23 to 25 March 2026. Project partners from across Europe came together for three days of presentations and discussions. Before the meeting I helped with the practical preparations, and during the three days I supported the team on site, helping participants and making sure that the sessions ran smoothly.



Lokesh Goud Dabbi
Thünen-Institut

Figure 29. At the AI4SoilHealth Consortium Meeting in Braunschweig (23 to 25 March 2026). Left: in front of the project banner. Right: my badge for the meeting.

Being part of the team also meant that I could follow the scientific programme. One of the talks, by Axel Don, gave an overview of soil carbon in Germany. It showed a map of predicted SOC stocks in agricultural soils in the upper 30 cm at 100 m resolution and compared mean stocks of mineral soils under cropland, grassland and permanent crops with those of organic soils. The large gap between mineral and organic soils on that slide was exactly the pattern I later found myself in the BZE data, where peat soils held about four times the carbon of mineral soils.

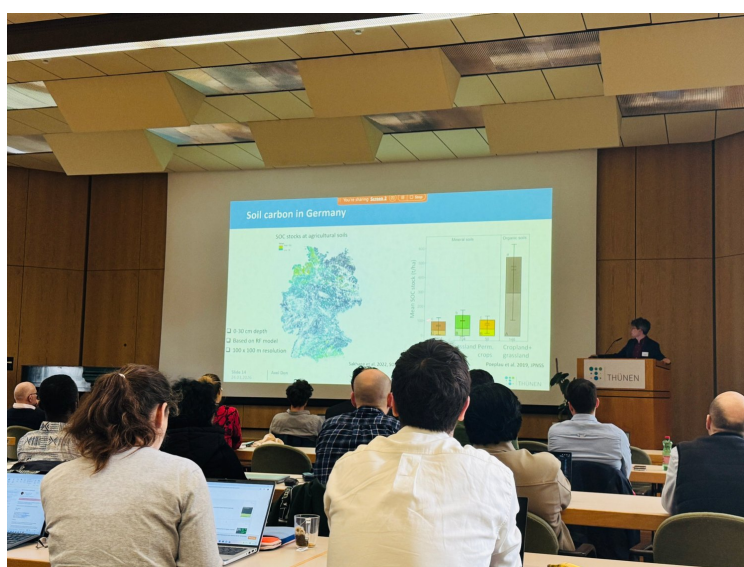


Figure 30. Presentation on soil carbon in Germany at the AI4SoilHealth Consortium Meeting. The slide shows mapped SOC stocks of agricultural soils (0 to 30 cm) and mean SOC stocks by land use, with organic soils far above all mineral soils.

Helping to run the meeting was a very good start to my internship. I got to know many colleagues in my first days, saw how a large European project is coordinated across many institutions and countries, and learned how much careful planning goes into a meeting of this size. The talks also showed me how data from programmes like BZE and LUCAS are combined with new methods such as machine learning, and how questions about soil health are linked to European policy.

7. Skills and Competencies Gained

The internship gave me a broad mix of scientific, technical and personal skills:

- **Soil carbon science:** a clear understanding of carbon stock and concentration, depth profiles, and the roles of climate, land use, texture, bulk density and peat.
- **Data analysis in R:** reproducible analysis pipelines with the tidyverse, terra and sf; cleaning and joining large tables, summary statistics, correlation analysis, and grouped comparisons across land use and climate classes.
- **GIS and spatial data:** collecting and processing spatial data in QGIS, and working with digitised, georeferenced aerial photographs from 1950 to the present.
- **Data visualisation:** publication ready figures, tables and descriptive statistics, from distributions and box plots to maps, used to tell a coherent story.
- **Building a dataset:** reconstructing land use history for about 1,000 LUCAS-SOIL sites, integrating it with published soil carbon data, classifying transitions and keeping a metadata table for reproducibility.
- **Project and event organisation:** planning, logistics and onsite support for the AI4SoilHealth Consortium Meeting, an international EU project meeting.
- **Scientific thinking:** recognising confounding factors such as climate, judging small samples carefully, and separating correlation from causation.
- **Communication:** discussing results with my supervisor, presenting findings and writing them up in a clear and structured way.
- **Working in a research institute:** organising my own time, asking for help at the right moment and working in an international team.

Beyond these specific skills, I became much more confident in working independently on an open research question, from defining what to look at to deciding when a result is solid enough to report.

8. Challenges and Solutions

The first challenge came almost straight away. Only a week after I started, I was helping to prepare and run the AI4SoilHealth Consortium Meeting, before I really knew the institute or my colleagues. I handled this by staying in close contact with the people organising the meeting, asking whenever I was unsure, and treating it as a chance to get to know the team quickly.

The next challenge was the size and structure of the data. The soil inventory is spread over three linked tables with thousands of rows each, and at first it was not obvious which table held which information

or how depth layers, horizons and sites belonged together. I solved this by drawing the structure of the tables for myself, checking every join with simple counts, and asking colleagues who had worked with the data before.

Another challenge was building the land use history. Examining historical aerial photographs in QGIS for about 1,000 sites takes time, image quality varies between years and countries, and for Spain the usable imagery basically ends around 2000. To keep the reconstruction consistent, I used the same land cover classes for every site, recorded the year and source of every observation in a separate metadata table, and only kept sites with either unchanged land use or one clear change.

A further challenge was scientific rather than technical. My first results suggested a very strong land use effect, but much of the afforestation in the dataset lies in cool northern Sweden, where soils are rich in carbon anyway. Only after discussing this with my supervisor did I start comparing land cover within temperature bands. This taught me to question results that look too clear, and to treat small groups, such as the 12 deforestation sites, with caution.

Finally, R itself was a learning curve. Producing figures that are both correct and easy to read took many attempts. Writing reusable scripts and getting regular feedback on my figures helped me improve quickly.

9. Reflection and Personal Development

One thing that happened during the internship really changed how I think about soil carbon research. When I first looked at soil carbon by land use, the differences between forest, grassland and cropland were so large and so consistent that it was tempting to see land use as the whole explanation. Only when I added climate to the analysis did the picture get more complicated. Warmer sites held much less carbon no matter how they were used. A simple comparison that ignored climate would have overstated the land use effect, because most of the afforestation in this dataset is in cool northern Sweden, where soils are rich in carbon because of the climate and not only because of the forest.

Realising this was an important lesson for me about method. Comparing land cover within separate temperature bands was useful exactly because it kept climate roughly constant and showed whether land use still made a difference, which it clearly did in every band. It showed me in a very practical way why observational studies on soil carbon have to control for confounding factors, and how a result that looks obvious in a simple plot can be misleading until the background variables are dealt with. It also made the difference between correlation and causation real for me, rather than something I had only heard about in lectures. The data could show consistent links, but proving cause would mean controlling for exactly the factors this analysis tried hard to separate, such as climate and the fact that certain transitions are concentrated in certain regions.

On a personal level, the internship gave me confidence in working at a research institute. I learned to plan my own work over several weeks, to present unfinished results without worrying too much, and to take critical feedback as part of the process. Working with colleagues from different backgrounds and taking part in seminars and meetings also helped me feel comfortable in a professional scientific environment in Germany.

10. Connection between Studies and the Internship

My master's programme in Organic Agriculture and Food Systems gave me the theory that this internship let me put into practice. Courses on soil science, soil fertility and the carbon and nutrient cycles taught me about soil organic matter, decomposition and the controls on soil carbon, and all of the results in this report build on that knowledge. Modules on sustainable land use and on agriculture and climate gave me the context to see why questions about afforestation, deforestation and protecting soil carbon matter, both scientifically and for society. During the internship I had to test ideas I had mostly met in theory against real measurements from thousands of sites.

The internship also improved my practical and analytical skills a lot. Working with the German Agricultural Soil Inventory and the LUCAS dataset showed me how large soil monitoring datasets are organised, how carbon stock and concentration are defined and measured, and how field, horizon and laboratory data are connected through site identifiers. Doing the whole analysis in R, from cleaning data and joining tables to calculating statistics and making figures good enough for publication, pushed my programming and statistics skills much further than any classroom exercise had. Building the land use reconstruction from historical aerial images was my first experience of creating a dataset from primary sources. It raised questions about consistency, uncertainty and documentation that simply do not come up when you work with data someone else has already prepared.

Apart from technical skills, the internship showed me how soil carbon research is actually done: how a research question becomes a dataset and an analysis, how figures are used to build an argument one step at a time, and how carefully results need to be qualified when working with observational data. Being part of the Regionalization and Data Science group at an institute responsible for national soil carbon reporting, and contributing to an EU project, also showed me how scientific analysis feeds into policy questions on land use and climate at national and European level. I want to work where soil science and data analysis meet, and this mix of soil knowledge and computational work is exactly the direction I want to go in. The internship made that goal clearer and stronger.

11. Conclusion and Outlook

In this report I have described my internship at the Institute of Climate-Smart Agriculture of the Johann Heinrich von Thünen Institute, where I carried out a research project on soil organic carbon in two phases. An exploratory analysis of the German Agricultural Soil Inventory helped me understand soil carbon data and what controls carbon storage. The main project then looked at how afforestation and deforestation affect topsoil organic carbon along a European climate gradient, using LUCAS soil data combined with my own reconstruction of land use history for Spain, France, Baden-Württemberg and Sweden.

In addition, I contributed to the EU project AI4SoilHealth and helped with the planning, logistics and onsite support of its consortium meeting in Braunschweig.

The main findings can be summarised as follows:

- In the German inventory, carbon storage depends on soil type, texture, structure and land use. Grassland always holds more carbon than cropland, and peat soils store about four times the carbon of mineral soils even though they make up fewer than 5% of sites.
- Across the European study area, soils with forest anywhere in their history hold about twice the carbon of cropland or grassland, and this forest signature lasts for decades after the land use change.
- Afforestation goes along with clearly higher topsoil carbon than the land without forest it replaced (about 45 compared with 18 g/kg). Deforestation shows the opposite, but the sample is small, so this result is tentative.
- Climate is the strongest single control on soil carbon, which drops sharply as mean annual temperature rises. Land cover still affects carbon independently of climate, as woodland, grassland and cropland keep the same order in every temperature band.
- Land use change follows regional patterns, with afforestation in Sweden and Spain and conversion between cropland and grassland in France and Spain. The recent carbon trend follows current land cover rather than past change, and woodland soils are still gaining carbon.

Taken together, these results support the main message from the exploratory analysis. Soil type and texture cannot be changed, but land use can, and it has a real effect on soil carbon. Protecting and expanding land cover that is rich in carbon, such as forest, grassland and above all peatland, is an effective way to protect soil carbon and, with it, the climate. On a personal level, the internship combined soil science and data analysis in exactly the way I hope to build my career, and it gave me both the practical skills and the scientific judgement needed to work with large soil carbon datasets. I would like to thank my supervisor Dr. Florian Schneider and the Institute of Climate-Smart Agriculture for giving me the opportunity to do this work.

Looking ahead, I would like to build on this experience in my master's thesis, ideally on a topic that combines soil carbon with data analysis, and to keep developing my skills in R and statistics. In the long term I see myself working in research or in an institution that uses soil data to support decisions on land use and climate.

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Appendix: Glossary of Terms

Term	Definition
Afforestation	Conversion of land that was not forest (cropland or grassland) into forest.
Deforestation	Clearing of forest and conversion of the land to another use.
Georeferencing	Assigning map coordinates to an image, such as an aerial photograph, so that it lines up with other spatial data.
GIS	Software for collecting, processing and analysing spatial data, such as QGIS.
Bulk density	Mass of dry soil per unit volume; used to convert carbon concentration into carbon stock.
Carbon concentration	Amount of organic carbon per mass of soil (g/kg).
Carbon stock	Total organic carbon stored under a given area to a given depth (Mg C/ha).
Confounding factor	A variable, such as climate, that influences both the cause and the outcome studied and can distort a simple comparison.
Mineral soil	Soil dominated by mineral particles with relatively little organic matter.
Peat (organic) soil	Soil formed under permanently wet conditions with a very high content of organic matter.
Persistent land use	Land use that did not change during the period studied.
Reproducible pipeline	A sequence of scripts that turns raw data into results and can be run again from the start to give the same output.
Topsoil	The upper soil layer, in this report mainly the upper 20 to 30 cm.
Land use transition	A change from one land use to another, such as from cropland to forest.